

The Effect of Teacher Feedback to Students' Question-asking in Large-sized Engineering Classes: A Perspective of Instructional Effectiveness and Efficiency

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This study compared the effects of teacher feedback and answers to the questions generated by students in a lecture-centered, large-sized class setting after considering the time the teacher spent providing feedback or answers. The following three research questions were addressed. (a) Are there any differences in the learning motivation level between students in the feedback and answer groups? (b) Are there any differences in cognitive engagement between students in the feedback and answer groups? (c) Are there any differences in achievement between students in the feedback and answer groups? The participants included 91 undergraduate students enrolled in a South Korean university. Feedback treatment and answer treatment were applied separately to two classes in the same undergraduate course titled "Structural Analysis I" for sixteen weeks of a semester. A survey and two achievement tests were conducted to assess the students' learning motivation, cognitive engagement, and achievement. This study revealed the teacher feedback to be better or equally effective on learning compared to the teacher answering the student's questions in a lecture-centered, large-sized class with more than 50 students after considering the teacher's time-on-task. Teacher feedback to the students' questions saves more time and is more critical than the teacher answering the students' questions. These findings suggest a practical instructional strategy in terms of enhancing the learners' cognitive engagement in large-sized engineering classes.

Keywords: Question-asking, teacher feedback, teacher answer, learning motivation, cognitive engagement, achievement

Expository lecture approaches are common instruction practices in higher engineering education institutes (de Jesus & Moreira, 2009), and are quite prevalent in many engineering colleges in South Korea. Engineering students need to understand and memorize key points of the materials presented to them during lectures so as to be successful academically in college. Learners tend to use a passive learning strategy, such as note-taking or re-reading because the teachers generally have control over teaching and learning for the class in the context of a teacher-led expository approach (King, 1992; O'Donnell, Reeve, & Smith, 2007).

Question-asking is recognized as one of the practical learning coping strategies that can be used within the limits of expository lectures (Dillon, 1988; Graesser & Olde, 2003; King, 1992, 1994; Pressley & Forrest-Pressley, 1985; van

der Meij, 1994). Large-sized expository lectures normally deter students from asking as many questions as they want (Fuhrer, 1989), and also prevent teachers from answering as many as questions they desire within the class hour. Consequently, students tend not to ask questions during class, which can lead to a decrease in the frequency of instructor interaction and feedback to the students, resulting in less active student involvement in the learning process (Spear, 1984; Stanley & Porte, 2002). This phenomenon is generally observed in engineering classes in South Korea with more than 50 students.

Because students' questions in a classroom are infrequent and unsophisticated (Dillon, 1988; Graesser & Person, 1994), teachers should make the students ask questions regularly in large-sized expository classes, and implement the appropriate teacher intervention to the

questions generated by students to improve the effect of question-asking by the student. Students want immediate answers to their questions, so teacher answers can be very helpful for learning (Zhang, Zhao, Zhou, & Nunamaker, 2004). On the other hand, answering all oral or written questions generated by students in a large-sized classroom is time-consuming and labor-intensive.

Even if teaching strategies, such as question-asking and teacher answers, improve students' learning, these may not be acceptable to the teachers because the strategies may require too much time and effort. Although most previous studies concentrated on examining the effectiveness of teaching strategies, few focused on the practicality of teaching strategies in terms of the teacher hours and classroom management constraints. According to Davis' (1989) technology acceptance model (TAM), the perceived ease of use as well as the perceived usefulness may determine an individual's intention to utilize a new technology. In this respect, in addition to the usefulness of a specific teaching strategy, the perceived efficiency might encourage the teacher to implement a new strategy in his or her teaching activities.

The assumption that underlies "questions" is that answers must follow (van der Meij, 1994). Therefore, it would be desirable if students' questions are answered by the teachers. Teachers, however, are unable to respond to all the students' questions due to the shortcomings of a large-sized classroom, as mentioned before. Therefore, this study examined whether teacher feedback, which requires less time than providing an answer to the student's question, can be a substitute for the teacher's answer in the perspective of instructional effectiveness and efficiency. Specifically, this study compared the effects of a teacher's feedback and teacher's answers to students' questions considering the time spent doing so.

Student question-asking

Student question-asking is a critical learning strategy to improve students' engagement, motivation, and achievement. In mainstream research on question-asking, student question-asking has attracted the most interest to enhance student question-asking skills. The rationale identified most frequently is that question-asking promotes students' active engagement in learning and achievement (Ciardiello, 1998; de Jesus & Moreira, 2009; King, 1989, 1992; Rosenshine, Meister, & Chapman, 1996; van der Meij, 1994). The process of question-asking can make students pay more attention to the lecture, facilitate organization of the lecture contents, and help students connect the lesson points and contents to their prior knowledge. In addition, question-asking has potential in terms of enhancing a student's motivation because in this approach, both information acquisition and knowledge construction are student-centered (Graesser & Olde, 2003). Question-asking also develops student's higher cognitive skills, such as self-

regulation, critical thinking, and metacognition (Ikuenobe, 2001; Na, Lee, & Teo, 2011). This is because students can identify what they do not know and ask questions that focus on improving their own knowledge deficits.

Question-asking commonly covers two different phases; student question-generating and student self-answering. The process of question-generating explains the improvements in the learner's comprehension beyond the information provided by the answers (Graesser & Olde, 2003). Therefore, it has been consistently demonstrated that students' answering their own questions enhances their comprehension more than answering instructor-posed questions (Davey & McBride, 1986; King, 1989).

On the other hand, students rarely ask questions in the current classroom environment (Dillon, 1988; Graesser & Person, 1994), where most of students have trouble identifying their own deficits (Hacker, Dunlosky, & Graesser, 1998). Recently, researchers in cognitive science and education have advocated learning environments that encourage the students to generate questions (Dillon, 1988; Pressley & Forrest-Pressley, 1985; van der Meij, 1994). In cases where student question-asking is provided as a regular activity in classes, the implementation of an appropriate teacher intervention to questions generated by the students is needed to improve the effect of student question-asking.

Teacher feedback

Teacher feedback, as a practical teaching strategy, reinforces appropriate learner behavior, allows the learners to realize how they are doing and extends learning opportunities to each student (Butler & Winne, 1995; Miller, 2002; O'Donnell et al., 2007). The most common formats of teacher feedback include (1) corrective feedback, which informs whether the learner's response is correct or incorrect; and (2) elaborative feedback, which explains why learner's response is correct or incorrect. Corrective feedback helps immediate learning, whereas elaborative feedback helps ensure deep understanding and ability to transfer knowledge (Kramarski & Zeichner, 2001; Moreno, 2004; Yang, 2011). Teacher feedback to questions generated by the students may encourage them to reflect on their questions (Kramarski & Dudai, 2009). In addition, if teacher feedback guides the students in how to generate questions, it may help students generate relevant, high level of cognitive questions, and enhance learning (Moreno, 2004). Nevertheless, most research focused on the effectiveness of students' self-question-asking, self-question-asking training, and teachers' question, particularly in the area of human social studies including science education. Few studies have examined how to give feedback to questions generated by the students.

Significance of the study

This study examined the effectiveness and efficiency of teacher feedback or answers to written questions generated

by the students. This study differs from previous studies, which focused on the effectiveness of the teaching strategies regardless of the teacher's time and effort. The present study collected time-on-task data for teacher feedback and answer activities to the questions generated by the students. Time-based variables, for example, the students' time allocated to learning, have been used to provide insight into the learning process and help explicate the learning results (Lin & Dwyer, 2010; Smyth, 1985). On the other hand, the time-on-task data in this study was used to evaluate the efficiency of a teaching strategy.

Both the teacher's feedback and answers to students' questions can be very helpful for learning. Questions require a response (Browne & Keeley, 2007). When students are provided with answers to their questions, they might be satisfied with the answers offered and have a successful experience by relieving their cognitive imbalance (Graesser & Olde, 2003). Teacher feedback through comments on a student's oral and written work can also enhance their learning (Black, Harrison, Lee, Marshall, & William, 2002). In the case of being provided with teacher feedback, the learners may still have a chance to reflect on the questions they generated and voluntarily make efforts to search for answers by reviewing the materials provided. Nevertheless, it is not easy to determine which one (teacher feedback or teacher answer) is more effective in a large-sized classroom environment. To answer this, the following three research questions were addressed:

1. Are there any differences in learning motivation level between students in the feedback and answer groups?
2. Are there any differences in student's cognitive engagement between students in the feedback and answer groups?
3. Are there any differences in students' achievement between students in the feedback and answer groups?

RESEARCH METHODS

Participants

A total of 91 undergraduate students (80 males, 11 females) enrolled in a class at a South Korean university, and a teaching faculty managing the class participated in this study. In the initial round of the survey, a total of 103 students (88 males, 15 females) participated. Those who had dropped out of the class and failed to take the tests were excluded. The students majoring in Civil Engineering took a prerequisite course titled "Structural analysis I". The teacher was a professor in the department of Civil Engineering with 13 years teaching experience. The experiment was carried out as part of the student's regular coursework in their class activities.

Study design

The group for research was split into two different classes: one class was assigned randomly to the feedback treatment, and the other, to the answer treatment. The same teacher taught the two classes. A pre-test was carried out to confirm the homogeneity of the two groups. Each student was required to write and submit one question at the end of each chapter to actively engage in their learning process or make good a deficiency in their knowledge of related topics. To provide elaborative feedback to the questions generated by the students in the feedback group, the teacher classified those questions depending on the cognitive level and informed each student of the cognitive level of each question. The teacher then shared two or three questions with a high cognitive level with the students and explained why those questions were relevant and valuable. For the answer group, the teacher wrote answer notes to the questions generated by the students and returned the question sheet with the answer to the students. In addition to question-asking, an explanation of fundamental engineering knowledge and examples, five quizzes during class-hours, midterm exam, and final exam were also conducted for both groups.

Feedback group. The "question sheet" was returned to each student with the cognitive level marked, and two or three questions with a high cognitive level were shared during the class and the reasons for being selected were explained.

Answer group. The "question sheet" was returned to each student with the answers written.

Characteristics of the "Structural Analysis I" course

The course, "Structural Analysis I", was designed to teach the fundamental concepts and methods for analyzing civil structures. Basic and principal theories are explained and some sample engineering problems are worked through by the instructor to help the students understand the lessons. Since this course is taken by approximately 100 junior Civil Engineering students in a semester, the students are allocated into two classes under the same curriculum. The grades for the students, however, are evaluated in total rather than in each separate class, and are determined by a midterm examination (35%), final examination (40%), attendance (10%), several quizzes (5%), and student participation in the class (10%). The student participation includes question-asking (7%).

Measures

In the current study, the following variables were measured: (a) student learning motivation; (b) student cognitive engagement (self-regulation, cognitive strategy use, and effort) for the class; (c) student achievement (perceived ability, test 1 of midterm examination, test 2 of final examination); and (d) teachers' time for giving feedback or answers. Except for the tests, the students' own

effort and teacher's time-on-task, all used a 1-7 response scale for the measures (strongly disagree/strongly agree). The questions focused on the structural analysis problems explained during the classes.

Student learning motivation. The student learning motivation was assessed with three measuring scales: course-specific motivation (eight items; $\alpha=.92$), teacher-specific motivation (6 items; $\alpha=.89$), and classmate-specific motivation (3 items; $\alpha=.92$), where α indicates the confidence level. This scale is a combination of previously-validated scales and newly-created scales for this study. The course-specific motivation scale was obtained from Dornyei (1994), which measures the students' motivation related to the teaching materials and learning tasks. Sample items of the scale are; *"this class interested me in some points"*, and *"the class was useful for me"*. The teacher-specific motivation scale was based on the items used previously by Clement, Dornyei, and Noels (1994), which measured the motivational impact of the teacher's personality, teaching style and practice. The sample items included *"the teacher was very passionate"*, and *"the teacher encouraged me to study in a self-directed way"*. The classmate-specific scale was developed by referring to the group-specific motivation of Dornyei (1994) because the class for the study was managed not in a group activity but in an expository lecture. The three items for this scale included *"compared to other classes taken, I am content with the classmates in this class"*, *"if I attend this type of lecture, I hope to study with very similar types of classmates"* and *"all the classmates are compatible and amiable"*.

Student cognitive engagement. The student cognitive engagement was assessed using four measuring scales; self-regulation (nine items; $\alpha=.87$), deep cognitive strategy (seven items; $\alpha=.86$), shallow cognitive strategy (4 items; $\alpha=.82$), and effort (1 item). This scale measure was based on the items used successfully by Miller, Greene, Montalvo, Ravindran, and Nichols (1996). Self-regulation was assessed by addressing to what extent the students were involved in planning, organizing study efforts, setting up goals and self-monitoring of progress (e.g., *"before a quiz or exam, I plan out how I will study the material"*, *"it is easy for me to establish goals for learning in this course"*). The seven items for the deep processing level focused mainly on those strategies geared towards understanding the contents better (e.g., *"when studying, I try to combine different pieces of information from the course materials in a new way"*, *"I draw pictures or diagrams to help me solve some of the problems"*), whereas the five shallow processing items focused mainly on strategies taken for rote-learning-memorization (e.g., *"I try to memorize the steps for solving problems presented in the text or in class"*, *"when I prepare for tests, I reread the notes that I took during the*

class and review the problems solved"). Effort was assessed with 1 item of multiple choice covering five alternatives: *"Extremely high"* to *"Extremely low"*.

Student achievement. At the initial stage of the study, student prior knowledge on structural analysis was assessed using a pre-test to identify the homogeneity of prior knowledge between the feedback group and the answer group. At the end of the study, the student achievement was evaluated with an eight-item scale for perceived ability ($\alpha=.82$), test 1, and 2. The perceived ability scale from a previous study by Greene and Miller (1996) required students to judge their own ability for the current structural analysis class themselves. The sample items for this scale were *"I am confident that I have the ability to understand the ideas taught in this course"*, *"my skills are weak compared to other students in this class"*. The contents of the pre-test, test 1, and 2 were developed by the second author who was the teacher in the structural analysis class.

Test 1. The midterm examination was carried out in the middle of the semester. All following three problems were asked: (1) question 1 asked the students to analyze and draw member force diagrams of a statically determinate frame structure; (2) question 2 aimed to determine the reactions and member forces of a statically determinate planar truss structure; and (3) the final question asked students to analyze and draw member force diagrams of a statically determinate beam structure and to calculate its deformation. The maximum score for questions 1, 2, and 3 was 30, 30, and 40, respectively.

Test 2. The final examination was conducted at the end of the semester to measure the students' understanding of the entire curriculum. Similar to test 1, the following three problems were asked: (1) question 1 asked the students to analyze and draw member force diagrams of a statically indeterminate frame structure; (2) question 2 was to analyze the member forces of a complex structure composed of beam and truss members; and (3) question 3 was to determine the absolute maximum member forces under moving loads over a simply supported beam structure. The level of test 2 for this study was relatively difficult compared to the normal tests. For this reason the mean score of the students for the final examination for this research was relatively low. The maximum score for each question was 50 so that the total score for test 2 was 150.

Covariate: time-on-task. The teacher's time-on-task was used as a control variable, and was defined as the average time the teacher spent scoring the cognitive level of the students' questions from the feedback group and in writing the answer to the students' questions from the answer group. The teacher's time-on-task was evaluated for the first and 8th question-asking activities. The amount of time was collected in minutes.

Procedure

Teacher guidance. The first author explained the research purpose and method to the teacher who was a second author for 40 minutes prior to study implementation. The teacher planned the lessons with a student question-asking activity and reflected it with 7% credit in the student evaluation. The first author was to answer any questions regarding study implementation from the teacher.

Pre-test. The student motivation and their cognitive engagement were measured by targeting one prerequisite course titled "Mechanics of Materials" that the two groups had taken previously. A test assessing the students' prior knowledge of the course subject titled "Structural Analysis I" was designed for this study. Student motivation was measured using the same scale used in the post survey in this study, and cognitive engagement was measured only by self-regulation. The results of the pre-test showed no significant difference in the students' learning motivation (feedback group: $M=4.69$, $SD=.75$, answer group: $M=4.41$, $SD=.78$, $t=1.454$), cognitive engagement (feedback group: $M=3.74$, $SD=.83$, answer group: $M=3.67$, $SD=.97$, $t=.298$), and pre-knowledge related to the course (feedback group: $M=8.84$, $SD=2.42$, answer group: $M=9.09$, $SD=2.55$, $t=-.465$) between the two groups. As a result, the two groups were almost identical from the perspective of the dependent variables.

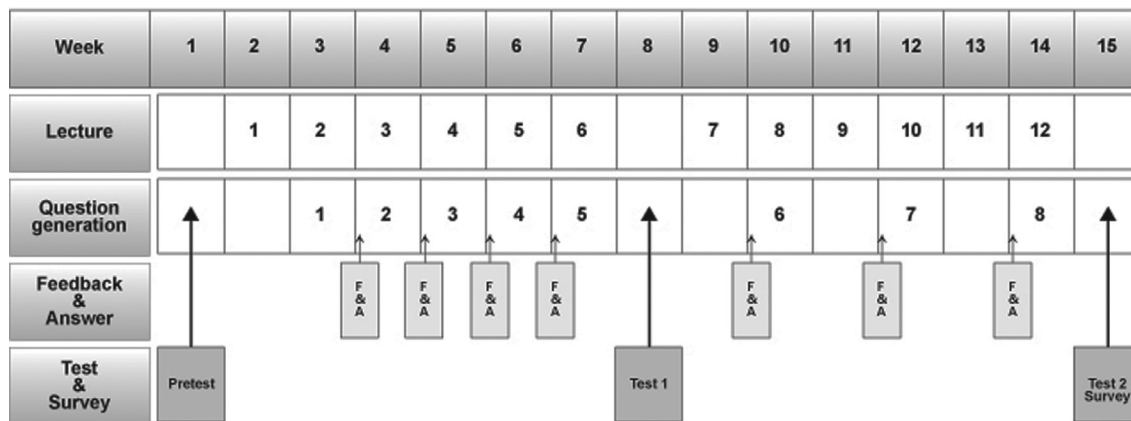
Study implementation. The students were instructed in two separate classrooms for a 16-week semester. All students were required to write down one question regarding the course taken upon completion of each chapter and submit it on a question sheet provided previously by the teacher. To reflect the students' question-asking activities to the students' evaluation, each question was analyzed independently and classified on the three-level scale reported by King (1994): low-level of factual questions, medium-level of comprehension questions, and high-level of integration questions. One point was scored for a factual question on the facts or other information covered explicitly in the lesson, 2 points for a comprehension question on a process or term to be described or defined, 3 points for an integration question on an explanation, inference, justification of what was not explicitly stated in the lesson, and 0 point for a question not related to the lecture. All students generated questions and those questions were analyzed eight times for this study. Among the questions, 9.24 % of these were not related to the lecture, 16.95% were factual questions, 69.19% were comprehension questions, and 4.62% were integration questions. Questions that students generated initially were used to examine the inter-rater reliability. Approximately 100 questions were analyzed by the teacher as an expert in structural analysis and a 98% inter-rater reliability was achieved in classifying the levels. For the feedback group, the teacher returned the question sheet to the student with its cognitive level, shared those questions that scored 3 points with the students, and explained why those questions were of good quality. For the answer group, the teacher provided

answers to each question. The teacher's time for providing feedback or answering was counted. It took approximately 1.0 minute for the teacher to read and classify each question according to the cognitive level for the feedback group, and approximately 5.3 minutes to write down the detailed answer to each question for the answer group.

One sample high-level integration question from the feedback group was: *"In the last class we learned about deflection. In the class, we just studied how to calculate deflections or slopes with the given loading conditions. On the other hand, in a real field, I think it is more important to know the load computation with given deflections. I saw a sign on the road saying that 'Trucks heavier than 20 tons cannot enter this bridge'. To know the weight of a truck we have to compute it reversely. I mean that in the class we compute deflection from given loads, but in the field sometimes we have to compute load based on given deflections. How can we calculate it in a reverse way?"* The teacher, the second author of the paper, noticed that this question is about an inverse problem and a highly sophisticated question. Therefore, he tried to explain the basic concept and importance of the inverse problems in engineering and science fields to the students, and also explained why this question should receive a high-level score.

Another example of a comprehensive question of a medium-level from the answer group was: *"Castigliano's 1st theorem can be extensively used even in nonlinear elastic structures. Then, why don't we use this theorem frequently?"* The teacher then wrote down an answer to this question and returned the question sheet to the student: *"Most structural problems can be solved based on the linear elastic assumption. It is more difficult to apply nonlinear approaches to analyze structural problems."*

Testing and surveying. The pre-test, test 1, and test 2 was conducted at the beginning of the study, in the middle of the semester, and at the end of the semester, respectively, for both groups. The teacher marked tests 1 and 2, whose items covered the fundamental concepts of the subject and consisted of objective items for applying problem-solving formulas. Since the course was to analyze a structural system by calculating the member forces and deflections due to the applied loads and by drawing member force diagrams, each test could be scored by the computed values and drawings in a very systematic manner. The teacher provided a guideline for scoring each test to the teaching assistant of the class, who marked every test item. After marking the scores of each test, the scores were open to the students so that any student could visit the teaching assistant or the teacher to ask for a re-evaluation of the mark. At the end of the semester, the grade of the course for each student was given by the teacher with all the data from the test results. Before conducting test 1, the students submitted a question sheet five times, and submitted one three times before test 2. A survey was carried out on the day of test 2 to check the other variables.



F: teacher's feedback to the students' questions

A: teacher's answer to the students' questions

Figure 1. Schematic diagram of the research design and procedure.

Analysis

The collected data was analyzed by multivariate analysis of covariance (MANCOVA) to determine the effects of teacher feedback and answers on the student learning motivation, cognitive engagement, and achievement adjusted according to the teacher's time-on-task. The dependent measures for the MANCOVA included the learning motivation (course-specific, teacher-specific, classmate-specific, and total learning motivation), cognitive engagement (self-regulation, deep cognitive strategy use, shallow cognitive strategy use, effort, and total cognitive engagement), and achievement (perceived ability, test 1, test 2). The teacher's time-on-task was used as a covariate. An independent sample *t*-test was used to determine if the treatment groups of the teacher feedback or teacher answer involved a different amount of time spent by the teacher. The results indicated a significant difference between the groups: feedback group: $M=89$, $SD=.09$, answer group: $M=5.29$, $SD=1.27$, $t=22.863$. The teacher spent significantly more time answering the questions from the answer group students than providing feedback to the feedback group students. Two prerequisites, homogeneity of variances and the correlations between the three dependent variables, were confirmed before performing the MANCOVA.

RESULTS

The results showed that the adjusted mean values for all dependent variables in the feedback group were higher than those in the answer group. Table 1 lists the descriptive statistics.

Difference in student motivation level between groups

Learning motivation survey consisted of three sub-scales, including course-specific, teacher-specific, and classmate-specific motivation. As shown in Table 1, the

entire mean values ranged from $M=4.83$ to $M=5.81$ out of 7.0, which indicates that the students in both groups possessed above average learning motivation. MANCOVA showed that the main effect of teacher feedback did not reach statistical significance for student learning motivation ($F(1, 89) = 2.397$, $p=0.125$, $\eta^2=.027$). Among the three sub-dependent variables of learning motivation, the result showed a significant effect of teacher feedback on course-specific motivation ($F(1, 89) = 5.310$, $p=0.024$, $\eta^2=.057$), but no significant effect on the teacher-specific ($F(1, 89) = 0.381$, $p=0.539$, $\eta^2=.004$) and classmate-specific ($F(1, 89) = 1.175$, $p=0.281$, $\eta^2=.013$) items.

Difference in student cognitive engagement level between groups

The main effect of teacher feedback was statistically significant for student cognitive engagement ($F(1, 89) = 16.643$, $p=0.000$, $\eta^2=.159$). Cognitive engagement variables consisted of four sub-variables, including self-regulation, deep cognitive strategy use, shallow cognitive strategy use, and efforts. The result of MANCOVA did reveal significant effects of teacher feedback to the student's question on their self-regulation, $F(1, 89) = 10.969$, $p=.001$, $\eta^2=.111$, deep cognitive strategy use, $F(1, 89) = 11.494$, $p=.001$, $\eta^2=.116$, shallow cognitive use, $F(1, 89) = 11.587$, $p=.001$, $\eta^2=.116$, and learning efforts, $F(1, 89) = 10.333$, $p=.002$, $\eta^2=.105$.

Difference in student achievement between groups

The students' achievement was assessed using three scales, including perceived ability, test 1, and test 2. MANCOVA revealed no significant difference in the effect of teacher feedback or answers on the perceived ability between the two groups ($F(1, 89) = 1.835$, $p=.179$, $\eta^2=.020$) and test 2 ($F(1, 89) = 2.673$, $p=.106$, $\eta^2=.029$), but a significant effect between the groups on test 1 ($F(1, 89) = 4.050$, $p=.047$, $\eta^2=.044$). The mean values of the students'

Table 1*Descriptive Statistics for the Dependent Variables of the Two Groups*

Dependent measures	Treatment			
	Feedback group (n=47)		Answer group (n=44)	
	<i>M</i> (<i>SD</i>)	Adjusted <i>M</i> ^a	<i>M</i> (<i>SD</i>)	Adjusted <i>M</i> ^a
<i>Time-on-Task</i>	0.89(0.09)	-	5.29(1.27)	-
<i>Learning Motivation</i>				
course-specific	4.99 (0.86)	5.48	4.83 (0.98)	4.30
teacher-specific	5.71 (0.88)	5.91	5.81 (0.88)	5.60
classmate-specific	5.26 (1.03)	5.63	5.39 (1.05)	4.99
Total	5.32 (0.79)	5.67	5.34 (0.85)	4.97
<i>Cognitive Engagement</i>				
self-regulation	4.80 (0.70)	5.28	4.45 (0.72)	3.99
deep cognitive	4.92 (0.78)	5.58	4.77 (0.88)	4.07
shallow cognitive	5.24 (0.88)	5.99	5.05 (1.02)	4.25
efforts ^b	3.64 (0.79)	4.25	3.45 (0.88)	2.80
Total	4.65 (0.63)	5.23	4.44 (0.76)	3.78
<i>Achievement</i>				
perceived ability	4.43 (0.86)	4.60	4.08 (0.84)	3.93
test 1 ^c	48.83 (18.79)	55.75	40.32 (21.50)	32.92
test 2 ^d	50.74 (37.84)	70.55	56.70 (39.24)	35.55

a: Adjusted mean using teacher's time-on task as a covariate.

b: The range of scores was 0-5.

c: The range of scores was 0-100.

d: The range of scores was 0-150.

* The range of scores for the other variables except b, c, and d was 0-7.

perceived ability was $M=4.43$ ($SD=0.86$) for the teacher feedback group and $M=4.08$ ($SD=0.84$) for the answer group, which indicates that the students in both groups perceived their achievement to be above the mean level. On the other hand, as shown in Table 1, both groups gained fairly low scores on test 2. The students in the feedback and answer group had a mean score of 33.83% and 38.68%, respectively.

DISCUSSION

Interpretations of the results

These results showed a significant effect of teacher feedback on student cognitive engagement but not much effect on student learning motivation and achievement. When students ask questions in the classroom, their teachers' answers are taken for granted. As a result, few studies have examined the effect of the teacher answers to students' questions in a classroom setting (van der Meij, 1994). The results of the present study in terms of the effect of teacher feedback on the students' activity were similar to those reported by Goldstein (2004), Konold, Miller, and Konold (2004), and Kramarski and Zeichner (2001).

Most classes in the department of Civil Engineering in a university are operated in the form of expository lectures with large-sized classes except for some design courses. Quizzes or tests are taken in most classroom lectures and indicate the students' learning progress through the scores achieved. On the other hand, few instructional strategies can cover feedback or comment on the individual learner's behavior or learning process in large-sized classes. Under these learning circumstances, both the feedback group and answer group were assessed as being relatively positive compared to the other normal courses. Consequently, the two groups demonstrated no significant difference in their teacher-specific and classmate-specific learning motivation. The course-specific learning motivation only revealed a significant effect on teacher feedback under the control of the teacher's time-on-task. This supports the finding of Graesser and Olde (2003) who reported the effect of the question-asking activity on enhancing learning motivation. The evaluative feedback to students' question might stimulate the students' sense of competition and consequently affect their course-specific learning motivation.

Questions are asked when the students are confronted with obstacles in understanding lectures or learning

materials, obvious gaps in knowledge, or discrepancies between prior knowledge and current learning contents (Graesser & Olde, 2003). In order for students to generate questions during a class, students need to pay attention to the lecture, focus on the important contents, analyze the learning contents, and relate it to prior knowledge. Therefore, students' question-generating and teacher feedback, which guides the students to ask further questions with a relatively higher cognitive level, made the students actively engage in learning. This supports the findings of King (1994) and de Jesus and Moreira (2009), who suggested that the formation of meaningful questions make the students actively engage in learning. With respect to student cognitive engagement, the feedback group showed better results than the answer group, even after considering the teacher's time spent providing feedback or answers. This observation explains certain meta-cognitive aspects of self-answering activities along with self-generating aspects. This result agrees with King's (1989) arguments in that the use of self-questioning as a metacognitive strategy made the students feel more in control of their own learning. When students are not capable of answering the question they generated, they need to take remedial action by asking related questions to themselves, posing the questions to their peers, or reading the materials and information sources provided (Brown & Palincsar, 1988; King, 1989). On the other hand, since the answer group has an expectation of direct answers from the instructor in the following class, few students appeared to have made efforts to search for answers on their own.

The answers to questions generated by the students may assist in immediate learning through filling a knowledge deficit because the student must determine what is not known when asking a question (Miyake & Norman, 1979). It is highly likely that some students in the feedback group perform additional cognitive activities to search for answers to the questions generated by them. Although the effect of teacher feedback or answers on students' achievement could not be determined, teacher feedback was found to be more effective in test 1, but there was no significant difference in their perceived ability and test 2 after considering the teacher efforts. Direct answers to the questions generated by students and self-answering may similarly affect their learning outcomes. Answers to the questions influenced by cognitive disequilibrium, regardless of them being teacher-led or students-led answers, restores the students' equilibrium and can affect the students' comprehension and knowledge of the subject matter (Graesser & Olde, 2003; Graesser & Person, 1994). On the other hand, there may have been less substantial difference observed through treatment, because the test items for test 2, which measures the students' achievement, were significantly more difficult than the students' level in this study. Therefore, careful discretion is advised when interpreting the results that compared the student achievement scores in the two groups.

Empirical contributions

Teaching and learning strategies should be effective and efficient for learning. Many studies have suggested effective teaching strategies in classroom environments but most of those researches overlooked the teacher's efforts required for implementation. This study examined the effect of the teacher feedback to the students' questions after controlling for the teacher's time for implementing it under the learning environment with the classroom management constraints of large-sized classes. Time-based variables have been used to predict the success in learning tasks, measure the students' engagement on learning, or examine the relationship with the learning outcomes (Lin & Dwyer, 2010). This study suggests that time-based variables can be used to measure the efficiency of a specific teaching strategy.

Theoretical contributions

Although some teaching strategies are quite effective for learning, they are not often implemented in teaching activities because they require too much of the teacher's time and effort, or the learning environments have some constraints for implementing them (El-Dinary & Schuder, 1993; Rich & Pressley, 1990). The teachers try to learn or implement the teaching strategies when they find them acceptable. The teacher's acceptance of teaching strategies is affected by their practicality and effectiveness (Rich & Pressley, 1990). This study examined empirically the acceptability of the teacher feedback strategy by examining the effect of teacher feedback that can be applied to large-sized classes by considering its efficiency.

CONCLUSION

This study examined how teacher feedback-and-answers in written format for questions generated by the students affect the learner's motivation, cognitive engagement, and learning achievements based on the teacher's time spent providing them. The following results were obtained.

In large-sized, lectured-centered classes with more than 50 students, a teacher's feedback has better or equally positive effects on learning than a teacher's answer to the students' questions. This suggests a useful strategy that can be applied to large-sized classes in the perspective of its effectiveness and efficiency. Specifically, providing feedback to students' questions is less time-consuming and more critical than providing answers to each of the students' questions.

In the case of teacher-led expository instructions with large-sized classes, providing detailed answers to the questions generated by the students is almost impossible in practice. Therefore, this study suggests a practical instructional strategy in terms of enhancing the learners' cognitive engagement in large-sized classes.

For more conclusive results, more study will be needed to determine the effects of question-asking targeted at

different types of lectures, such as design courses or practical courses.

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